INCH-POUND
MIL-DTL-16377/27D(SH)
26 May 2023
SUPERSEDING
MIL-DTL-16377/27C(SH)
30 June 2014

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, GENERAL LIGHTING, 50 WATTS, 120 VOLTS, WATERTIGHT, SYMBOLS 92.2, 93.2, 93.4, 93.5, 92.2SSL, 93.2SSL, 93.4SSL, AND 93.5SSL

This specification is approved for use by the Naval Sea Systems Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 11 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.



FIGURE 1. Typical view of symbols 92.2 and 93.2.

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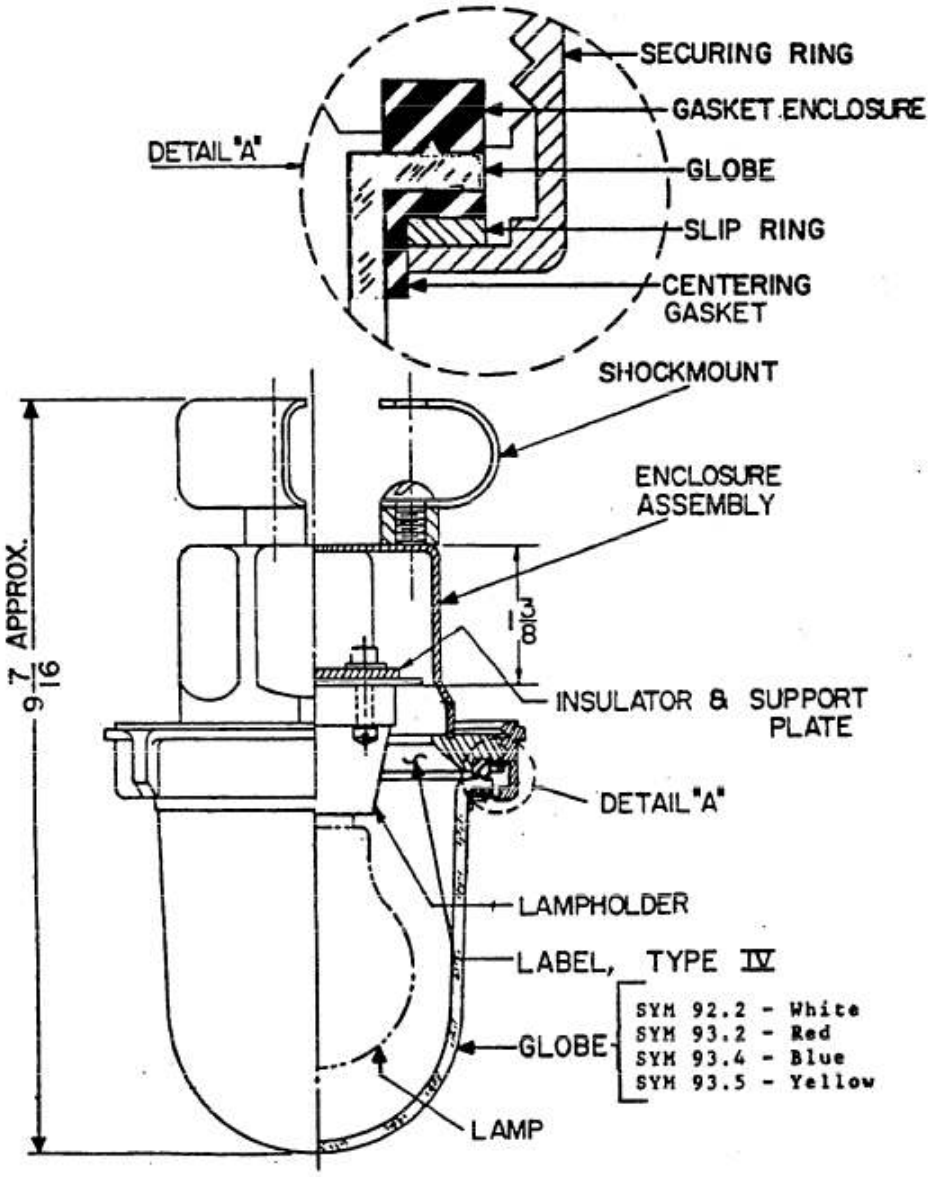


FIGURE 2. Fixture assembly.

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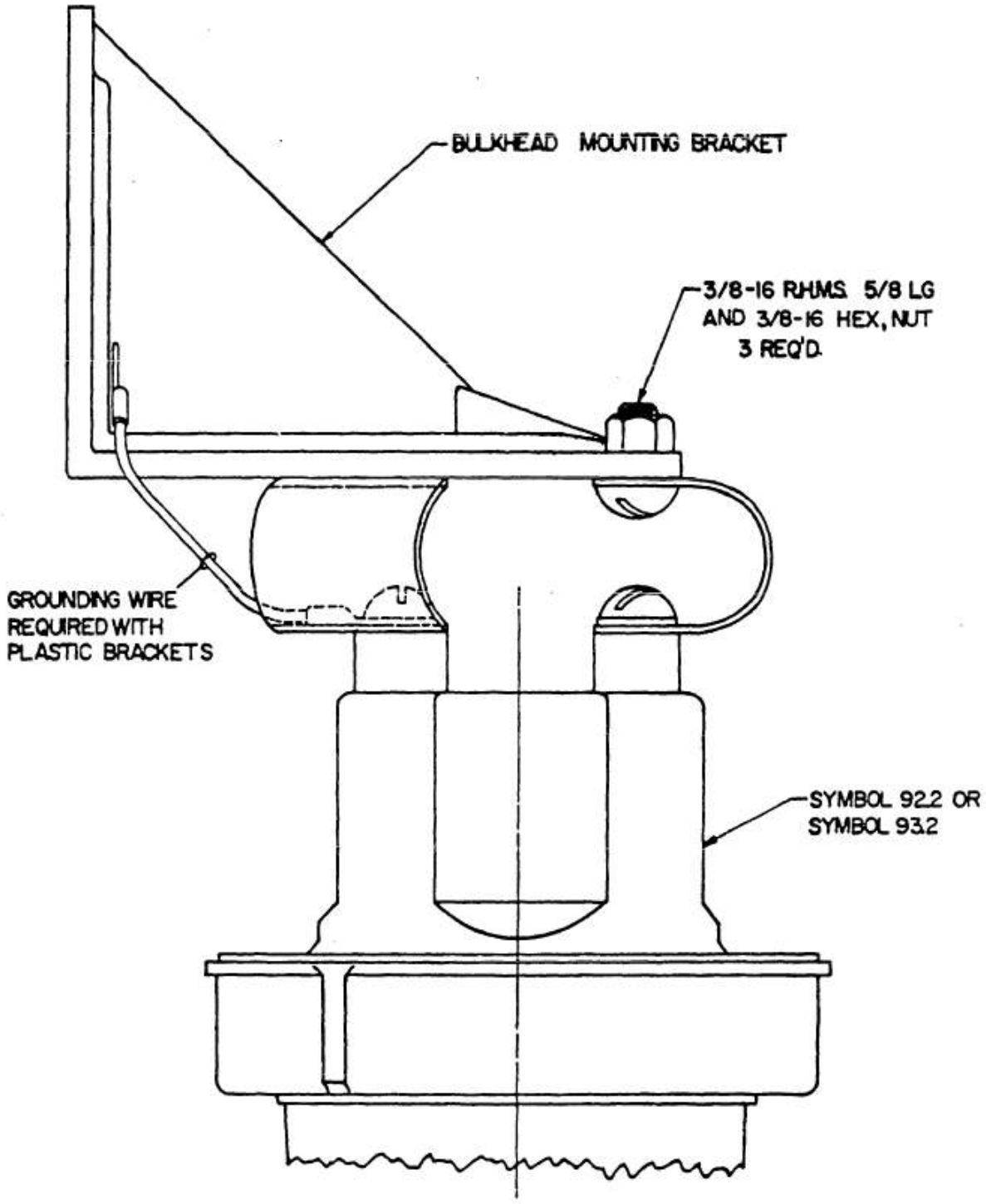


FIGURE 3. Fixture assembly for bulkhead mounting.

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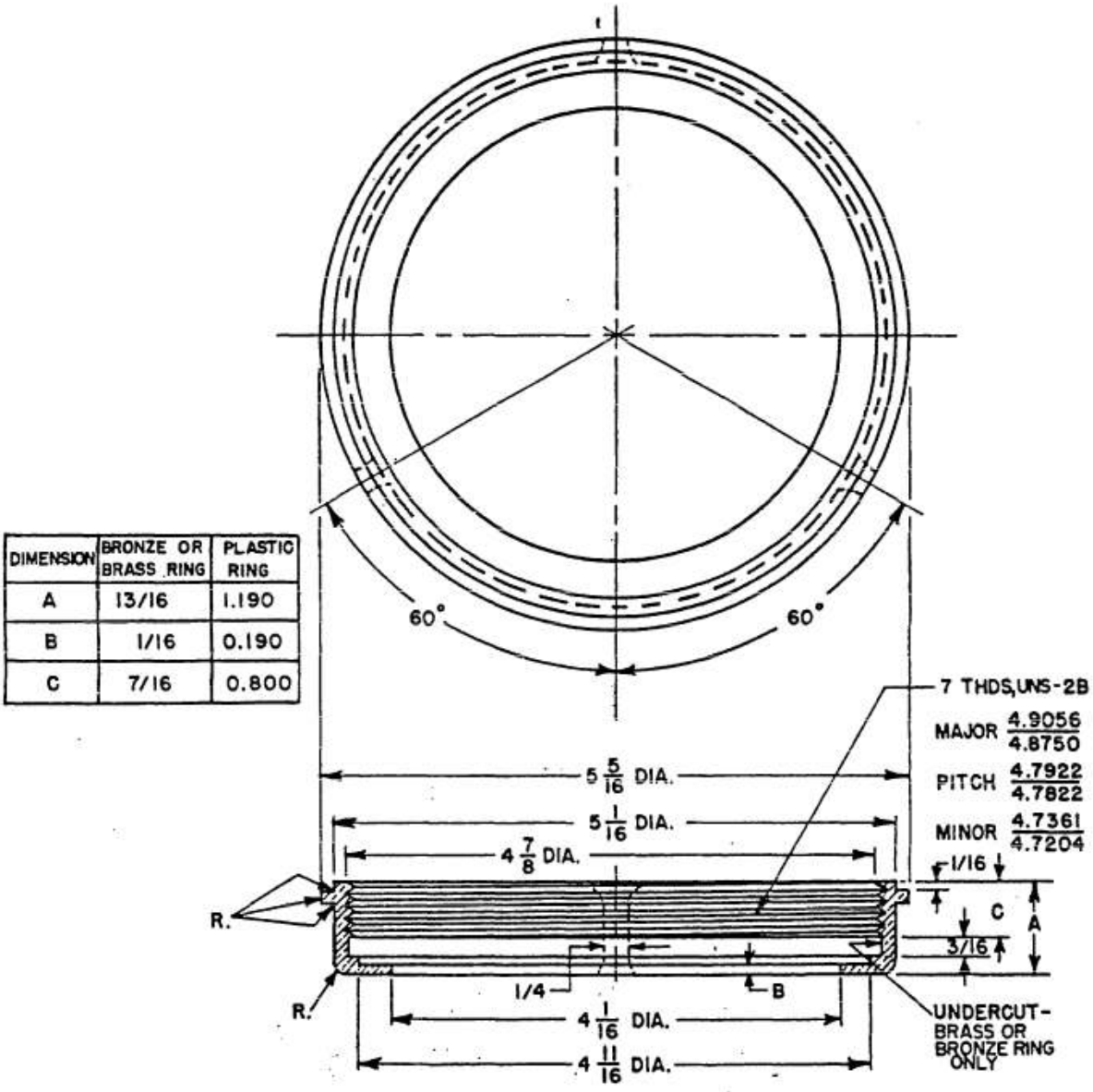


FIGURE 4. Securing ring.

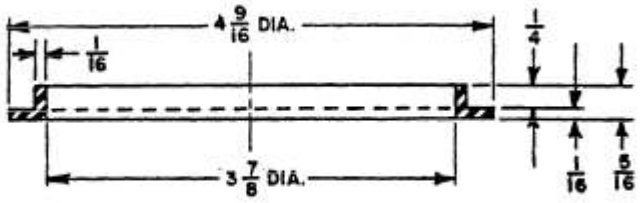


FIGURE 5. Globe centering gasket.

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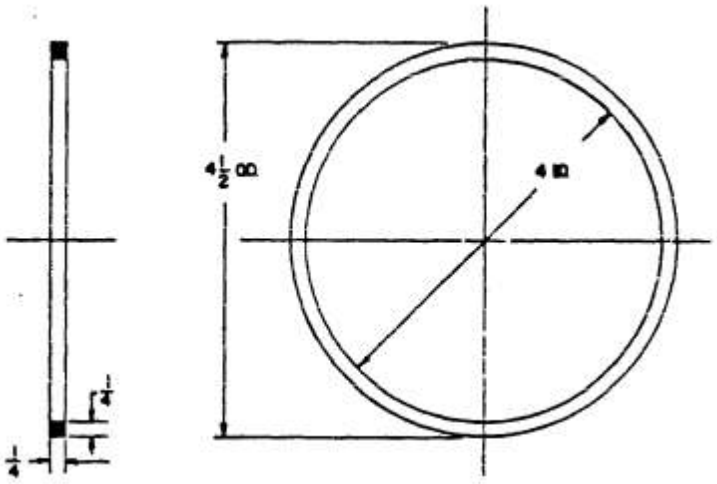


FIGURE 6. Enclosure gasket.

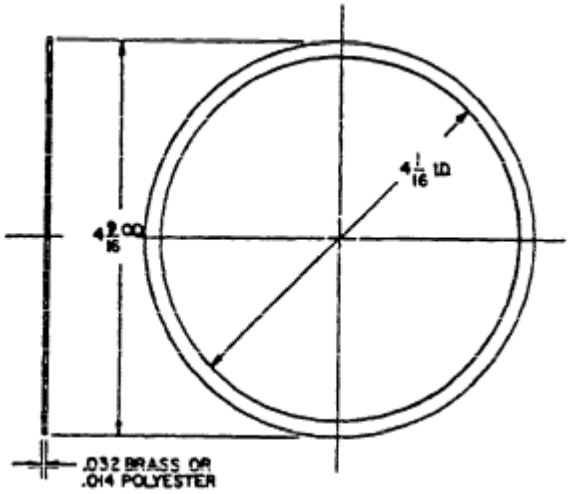


FIGURE 7. Slip ring.

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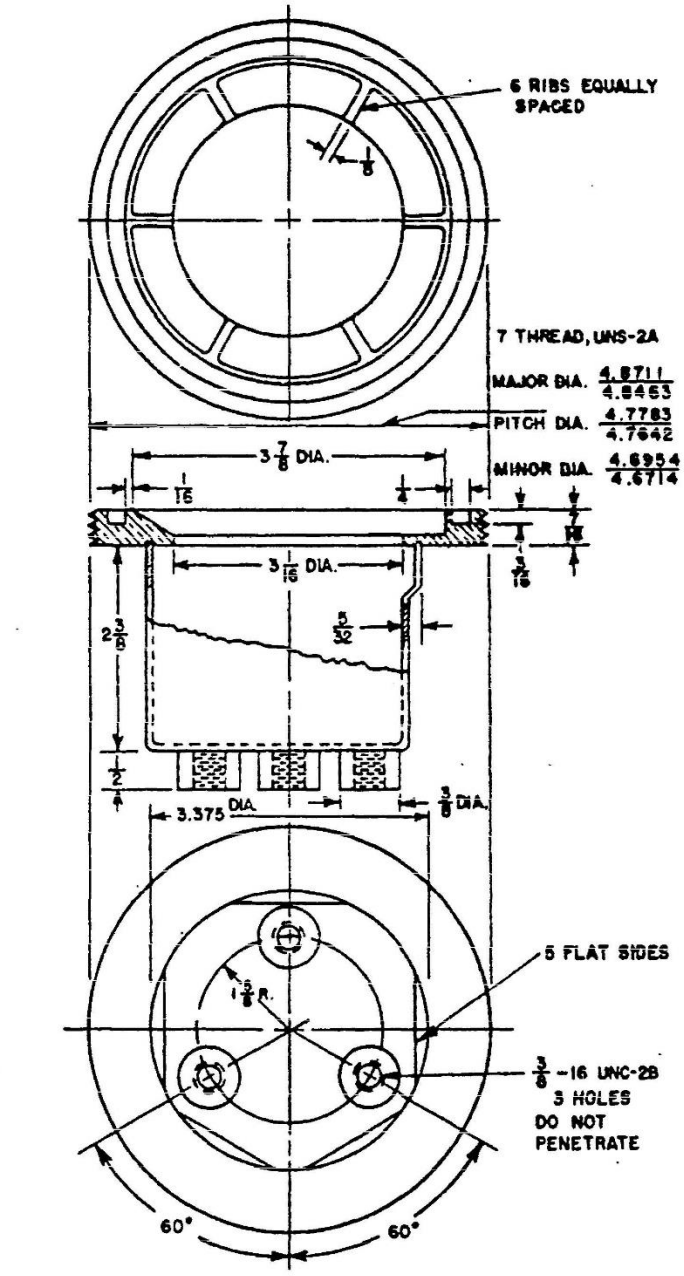
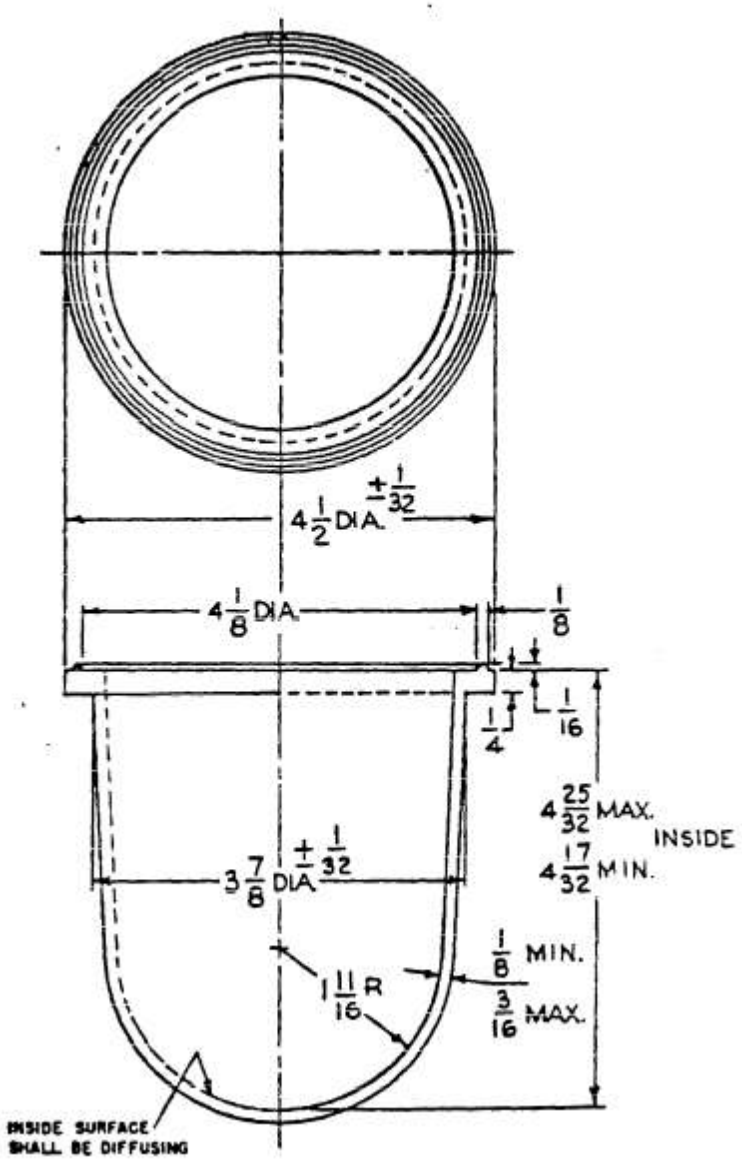


FIGURE 8. Enclosure assembly.

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Material: Tempered glass

Color:

Symbol 92.2: Clear transparent/diffusing

Symbol 93.2: Red transparent/diffusing

Symbol 93.4: Blue transparent/diffusing

Symbol 93.5: Yellow transparent/diffusing

Transmittance (symbol 92.2): Clear transparent/diffusing – 80 percent minimum

FIGURE 9. Globe.

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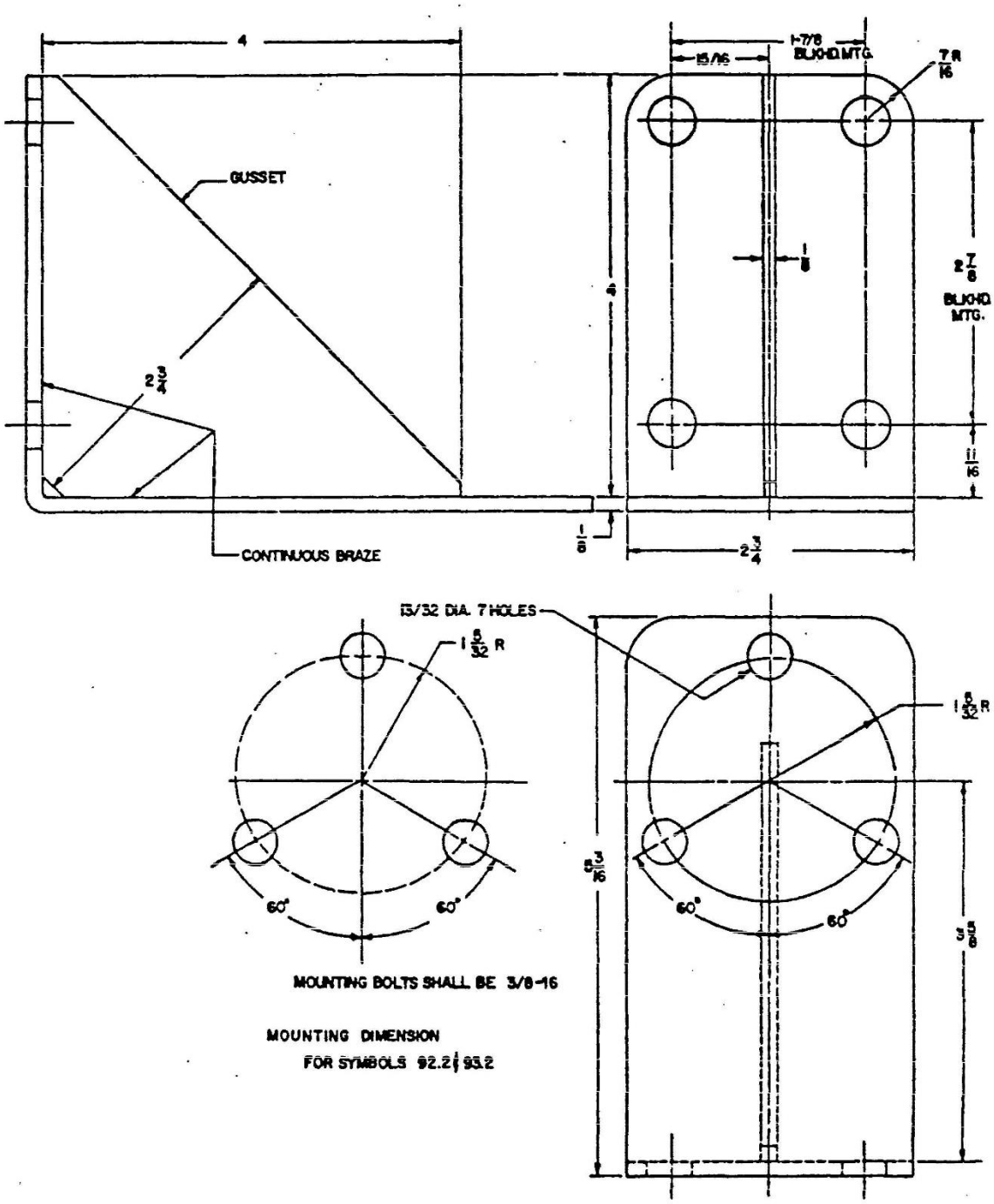
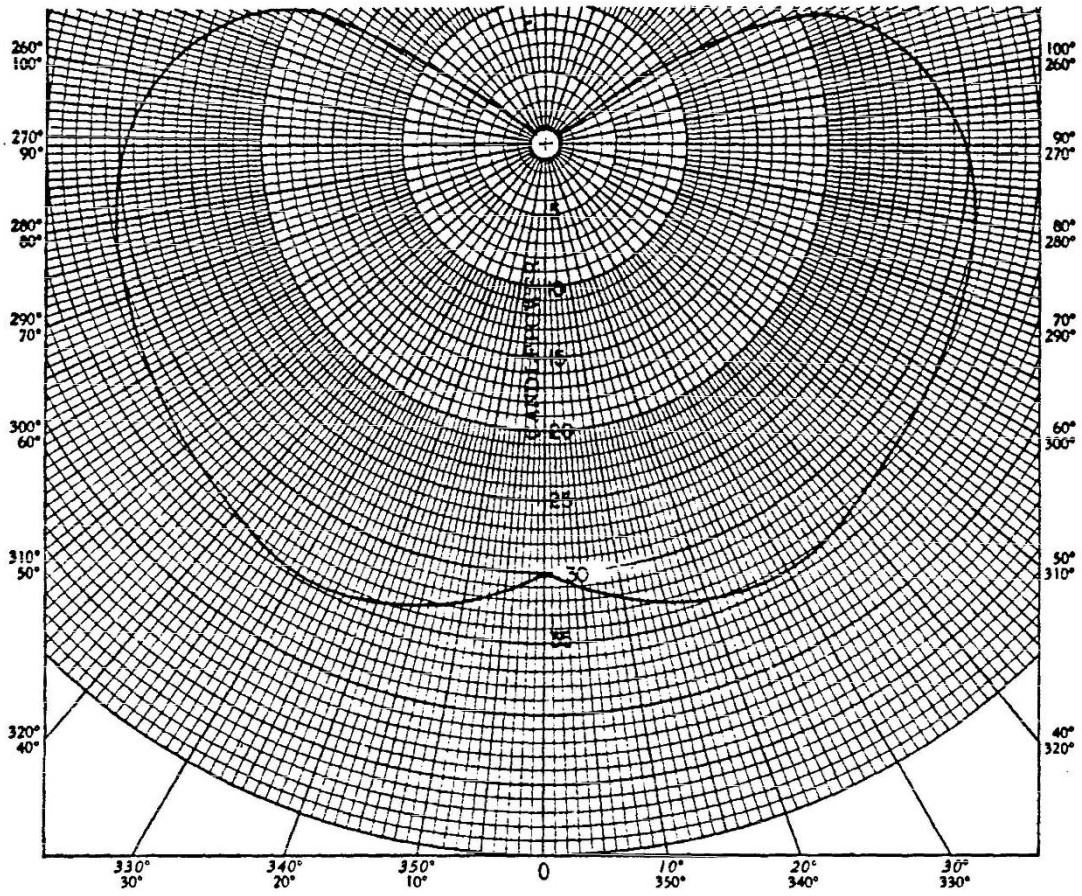


FIGURE 10. Mounting dimensions and bracket for bulkhead mounting (metal bracket only).

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ANGLE	CANDLEPOWER
	SYMBOL 92.2
0	30
5	32
10	32.5
15	33
20	33.5
25	34
30	34
35	34
40	33
45	32
50	32
55	32
60	31.5

ANGLE	CANDLEPOWER
	SYMBOL 92.2
65	31
70	31
75	31
80	30.5
85	30
90	30
95	28
100	27.5
105	27
110	25
115	23
120	12.5
125	2

Photometric test data:

Distance: 10 feet

Voltage: 120 volts

Ambient temperature: 76 °F (24.4 °C)

Lamps: 50 watts, rough service conforming to A-A-50449, furnished by laboratory for test, shall be used

FIGURE 11. Candlepower distribution curve.

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INCANDESCENT REQUIREMENTS

CLASSIFICATION

Type: II
Class: 2

SYMBOL NUMBERS

92.2: Complete lighting fixture assembly, standard mounting, with one 50-watt white incandescent lamp and white (clear transparent) diffusing globe.
93.2: Complete lighting fixture assembly, standard mounting, with one 50-watt white incandescent lamp and red (transparent diffusing) globe.
93.4: Complete lighting fixture assembly, standard mounting, with one 50-watt white incandescent lamp and blue (transparent diffusing) globe.
93.5: Complete lighting fixture assembly, standard mounting, with one 50-watt white incandescent lamp and yellow (transparent diffusing) globe.

DESIGN

Electrical rating:	50 watts, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 2 through 9.
Enclosure assembly (includes box, threaded flange, tapped pads, and support plate):	See figure 8. Material shall be brass.
Insulator for brass box assembly only:	Plastic.
Gasket:	Rubber, ¼-inch-square cross section by 4-inch inside diameter (id) by 4½-inch outside diameter (od).
Lampholder:	UL 496, Supplement SB, plastic screw type, two-contact.
Shock mounts:	Brass, MIL-F-16377/41-012 or MIL-F-16377/41-015.
Centering gasket:	Molded rubber, see figure 5.
Globes:	
Symbol 92.2:	White (clear transparent) diffusing, see figure 9.
Symbol 93.2:	Red transparent diffusing, see figure 9.
Symbol 93.4:	Blue transparent diffusing, see figure 9.
Symbol 93.5:	Yellow transparent diffusing, see figure 9.
Slip ring:	Brass, 4⅞-inch od by 4⅛-inch id by 1/32 inch thick.
Securing ring:	Cast brass, bronze, or gray plastic, type MAI-60 in accordance with ASTM D5948.
Bulkhead mounting bracket:	When specified, a brass, corrosion-resistant steel. The metal brackets shall be as shown on figure 3. Thickness of material shall be as necessary to meet the requirements specified herein.
Lamp:	Fixtures shall accommodate a 50-watt white lamp conforming to A-A-50449. (Lamp not furnished.)
Distribution curves (symbol 92.2):	Candlepower (see figure 11).
Weight (symbols 92.2 and 93.2):	2 pounds, 13 ounces maximum.

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Designation and marking:	One type IV label shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figure 2. The information to be shown on the type IV label shall be: "INSTALL 50-WATT LAMP ONLY. LAMP: NSN 6240-00-143-3070 WHITE GLOBE: NSN 6210-00-702-5773 RED GLOBE: NSN 6210-00-914-4152"
Identification plate:	Brass, stamped or engraved.

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type:	III
Class:	2

SYMBOL NUMBERS

92.2SSL:	Complete lighting fixture assembly, general lighting, with LED light source and white (clear transparent) diffusing globe.
93.2SSL:	Complete lighting fixture assembly, general lighting, with LED light source and red transparent diffusing globe.
93.4SSL:	Complete lighting fixture assembly, general lighting, with LED light source and blue transparent diffusing globe.
93.5SSL:	Complete lighting fixture assembly, general lighting, with LED light source and amber transparent diffusing globe.

DESIGN

Electrical rating:	25 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 2 through 9.
Photometric data:	See figure 11 and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Mounting dimensions:	See figure 10.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Globes:	
Symbol 92.2SSL:	White (clear transparent), see figure 9.
Symbol 93.2SSL:	Red, see figure 9.
Symbol 93.4SSL:	Blue, see figure 9.
Symbol 93.5SSL:	Yellow, see figure 9.
Weight:	5 pounds, 10 ounces maximum.
Designation and marking:	Labels shall be furnished in accordance with MIL-DTL-16377 and the following: (a) One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by: (1) A list of all standard parts shown in table IV and globes of table IV by name and stock number only, if appropriate, or (2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer's part numbers.

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EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I for type I and table II for type III.
Globes:	See table III.
Conformance inspection:	
Comparison inspection:	Same as first article inspection.
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

(a) Assignment of military part numbers and general information is shown in table IV.

(b) Fixtures may be adapted for bulkhead mounting by using the bracket shown on figure 3.

TABLE I. First article and comparison inspection for complete type I fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	---
Operation	3.13	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14	1/ Watertight
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14	1/ Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Continuity of grounding	3.6.13.4	4.8.12	---
Optical uniformity	3.7.7.1	4.8.4	---
Mechanical shock	3.7.5.6	4.8.17.3	8 foot-pounds
Thermal shock	3.7.5.7	4.8.17.4	---
Breakage pattern	3.7.5.9	4.8.17.6	---
Color and luminous transmittance	3.7.3	4.8.6.3	Symbols 93.2, 93.4, and 93.5 only
Light output	3.7.1	4.8.6.1	Symbol 92.2 only Candlepower
Magnetic permeability	3.4.8	4.8.18	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE II. First article and comparison inspection for type III luminaires.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	---
Operation	3.13	4.8.1	---
Power interface	3.6.18	4.8.21	---
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14	1/ Watertight
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14	1/ Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Noise	3.5.14	4.8.9	1/
Continuity of grounding	3.6.13.4	4.8.12	---
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.17	4.8.24	---
Optical uniformity	3.7.7.1	4.8.4	---
Mechanical shock	3.7.7.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.7.7	4.8.17.4	---
Breakage pattern	3.7.7.9	4.8.17.6	---
Color and luminous transmittance	3.7.3	4.8.6.3	Symbols 93.2SSL, 93.4SSL, and 93.5SSL only
Stress relief	3.8.5.3	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---
Salt spray	3.5.15	4.8.10	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE III. First article and comparison inspection for globes.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.5.1	4.8.4	---
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Light output	3.7.1	4.8.6.1	Symbol 92.2 or 92.2SSL only Candlepower
Mechanical shock	3.7.5.6	4.8.17.3	8 foot-pounds
Thermal shock	3.7.5.7	4.8.17.4	---
Breakage pattern	3.7.5.9	4.8.17.6	---
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 93.2 only

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TABLE IV. Part number and general information.

Military Part Number M16377/27-	Item Name	National Stock Number (NSN)	Supersedes Part Number
92.2	Complete fixture, symbol 92.2 consisting of: Body White (clear transparent) diffusing globe	 6210-00-299-7703 1/ 6210-00-702-5773	 9000-S6401-74218 Symbol 92.2
93.2	Complete fixture, symbol 93.2 consisting of: Body Red transparent diffusing globe	 6210-00-299-7703 1/ 6210-00-914-4152	 9000-S6401-74218 Symbol 93.2
93.4	Complete fixture, symbol 93.4 consisting of: Body Blue transparent diffusing globe	 6210-00-299-7703 1/ 6210-01-042-8652	 ---
93.5	Complete fixture, symbol 93.5 consisting of: Body Yellow transparent diffusing globe	 6210-00-299-7703 1/ 6210-01-166-7187	 ---
92.2SSL	Complete fixture, solid state, symbol 92.2SSL	---	---
93.2SSL	Complete fixture, solid state, symbol 93.2SSL	---	---
93.4SSL	Complete fixture, solid state, symbol 93.4SSL	---	---
93.5SSL	Complete fixture, solid state, symbol 93.5SSL	---	---
001	White (transparent/diffusing) globe	6210-00-702-5773	MS17802-20
002	Red (transparent/diffusing) globe	6210-00-914-4152	---
003	Mounting bracket	6210-01-365-3635	---
004	Body (complete less globe and mounting bracket)	6210-00-299-7703 1/	---
005	Securing ring (legacy incandescent only)	6210-00-399-6207	---
006	Blue (transparent/diffusing) globe	6210-01-042-8652	---
007	Yellow (transparent/diffusing) globe	6210-01-220-3315	---

NOTE:

1/ Body material shall be brass in accordance with requirements section of this specification sheet.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 6210-2023-010)

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